

2026.01.15 CrossFreeze消融

消融实验设计

fedavg基线

```
python main.py \  
  --experiment baseline \  
  --dataset cifar10 \  
  --gpu 0 \  
  --seed 2024 \  
  --epochs 200 \  
  --num_clients 20 \  
  --frac 1.0 \  
  --local_epochs 5 \  
  --batch_size 64 \  
  --lr 0.02 \  
  --weight_decay 1e-4 \  
  --iid 0 \  
  --alpha 0.5 \  
  --save_dir ./results_cifar10_fedavg
```

Baseline - 各类别测试准确率摘要

类别	准确率
airplane	77.28%
automobile	86.97%
bird	72.72%
cat	65.98%
deer	70.94%
dog	67.79%
frog	78.58%
horse	85.52%
ship	87.43%
truck	85.28%
平均	77.85%

指标已保存到: ./results_cifar10_fedavg/baseline_fedavg_cifar10_noniid_2026-01-19_00-22-25/metrics.json

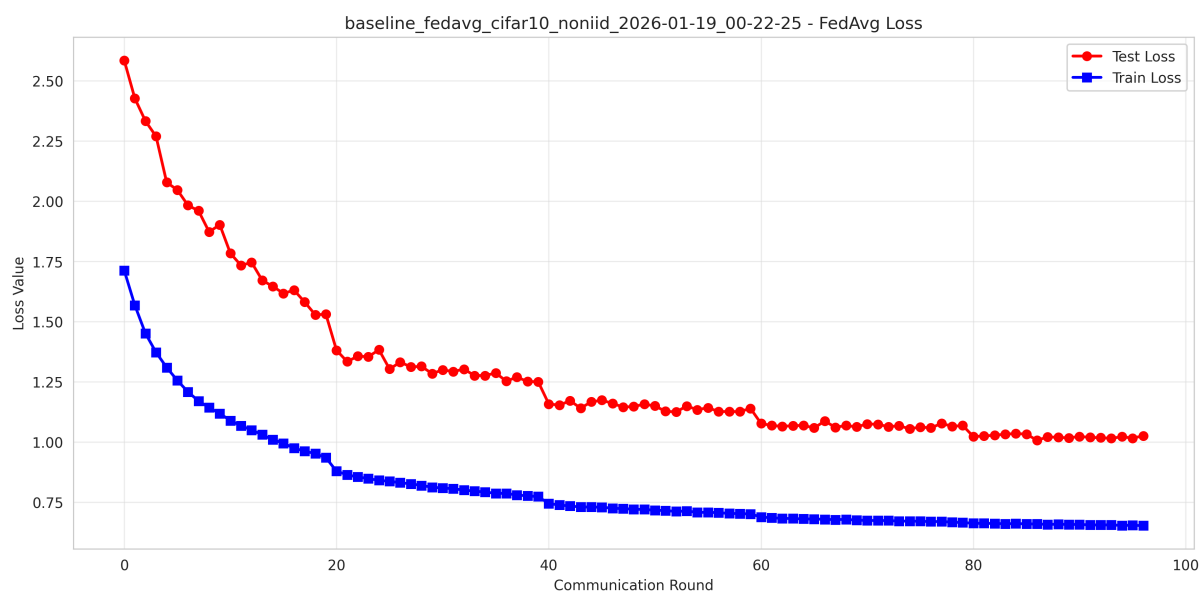
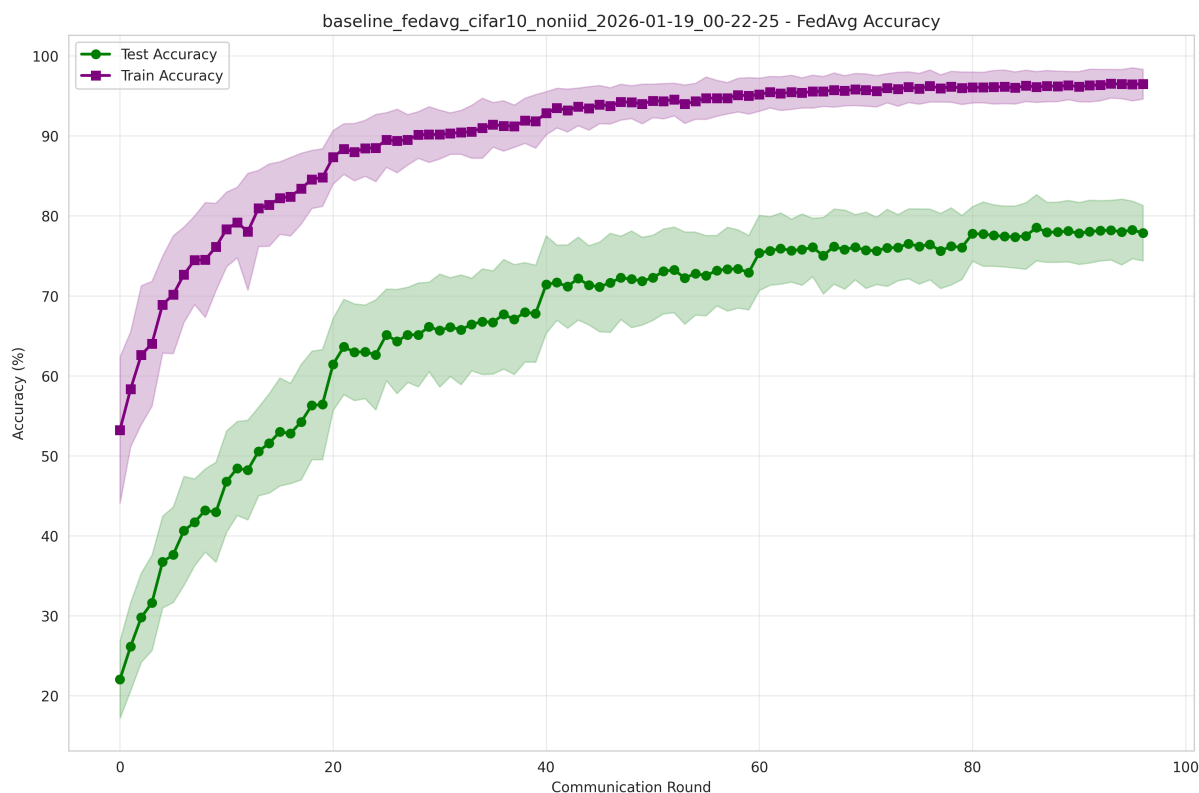
=== Baseline (FedAvg) Summary ===

最终测试准确率: 77.85%

最佳测试准确率: 78.52%

生成 Baseline 可视化图表...

Baseline 可视化图表已保存到: ./results_cifar10_fedavg/baseline_fedavg_cifar10_noniid_2026-01-19_00-22-25



A

配置含义: 仅使用基础增强, 仅优化本地 M2 损失, 无全局 L_{sm} 约束, 无一致性。**修改点:** `gamma_sm=0` (去除全局约束), `consistency_beta=0` (去除一致性), `use_cutmix=0` & `cutmix_alpha=0` (强制弱增强)。

```
python main.py \
  --experiment crossfreeze \
  --dataset cifar10 \
  --gpu 0 \
  --seed 2024 \
  --epochs 200 \
  --num_clients 20 \
  --frac 1.0 \
  --local_epochs 5 \
  --batch_size 64 \
  --lr 0.02 \
  --weight_decay 1e-4 \
  --iid 0 \
  --alpha 0.5 \
  --min_lr 0.002 \
  --gamma_sm 0.0 \
  --consistency_beta 0.0 \
  --use_cutmix 0 \
  --cutmix_alpha 0 \
  --separate_loss 0 \
  --save_dir ./results_cifar10_exp_A
```

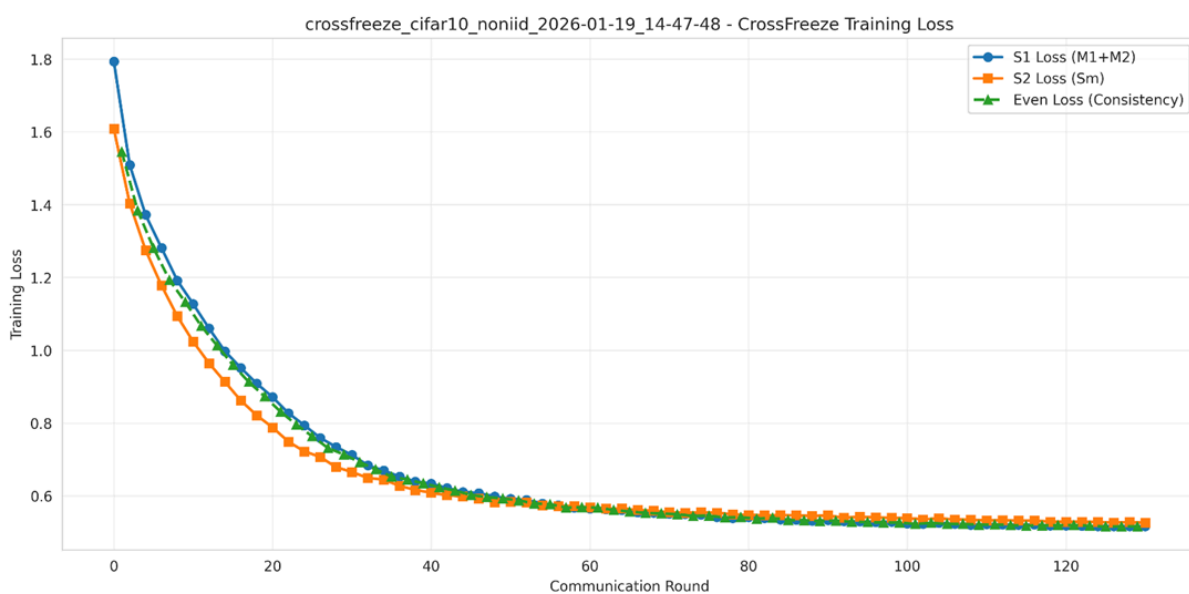
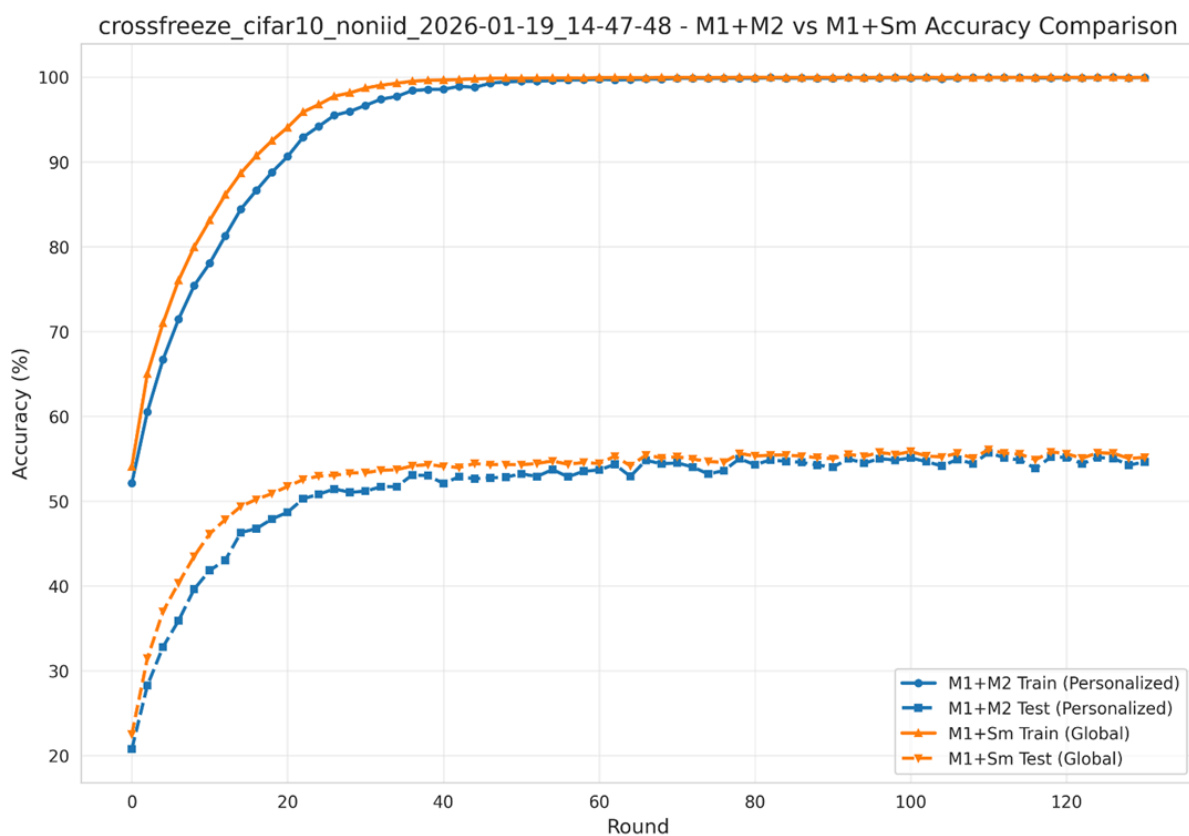
```

🔴 早停触发!
  当前准确率: 55.15%
  最佳准确率: 56.08% (第 56 轮)
  已连续 10 轮无改善 (阈值: 0.01%)
  提前结束训练 (计划 200 轮, 实际 131 轮)
训练进度: 65% | 130/200 [5:18:04<2:51:16, 146.80s/it]

=====
CrossFreeze - 各类别测试准确率摘要
=====
类别      M1+M2(个性化)  M1+Sm(全局)  差值 (M2-Sm)
-----
airplane   54.98%        55.17%       -0.19%
automobile 67.65%        67.98%       -0.33%
bird       49.77%        49.89%       -0.12%
cat        43.36%        44.08%       -0.72%
deer       37.68%        39.23%       -1.56%
dog        50.04%        50.65%       -0.61%
frog       58.56%        58.66%       -0.10%
horse      63.80%        64.50%       -0.70%
ship       60.74%        61.40%       -0.65%
truck      59.50%        59.92%       -0.41%
平均       54.61%        55.15%       -0.54%
=====

指标已保存到: ./results_cifar10_exp_A/crossfreeze_cifar10_noniid_2026-01-19_14-47-48/metrics.json
=== CrossFreeze Summary ===
M1+M2 最终准确率: 54.61%
M1+Sm 最终准确率: 55.15%
M1+M2 最佳准确率: 55.71%
M1+Sm 最佳准确率: 56.08%
生成 CrossFreeze 可视化图表...
CrossFreeze 可视化图表已保存到: ./results_cifar10_exp_A/crossfreeze_cifar10_noniid_2026-01-19_14-47-48
Accuracy curves saved: ./results_cifar10_exp_A/crossfreeze_cifar10_noniid_2026-01-19_14-47-48/accuracy_comparison.png

```



B

配置含义: 基础增强 + 双重损失 ($L_{m2} + L_{sm}$)，但使用静态权重。**修改点:**

`gamma_sm=2.0` (开启全局约束), `gamma_scheduler=static`, `use_cutmix=0` & `cutmix_alpha=0` (保持弱增强)。

```
python main.py \
  --experiment crossfreeze \
  --dataset cifar10 \
  --gpu 0 \
  --seed 2024 \
  --epochs 200 \
  --num_clients 20 \
  --frac 1.0 \
  --local_epochs 5 \
  --batch_size 64 \
  --lr 0.02 \
  --weight_decay 1e-4 \
  --iid 0 \
  --alpha 0.5 \
  --min_lr 0.002 \
  --gamma_sm 2.0 \
  --gamma_scheduler static \
  --consistency_beta 0.0 \
  --use_cutmix 0 \
  --cutmix_alpha 0 \
  --separate_loss 1 \
  --save_dir ./results_cifar10_exp_B
```

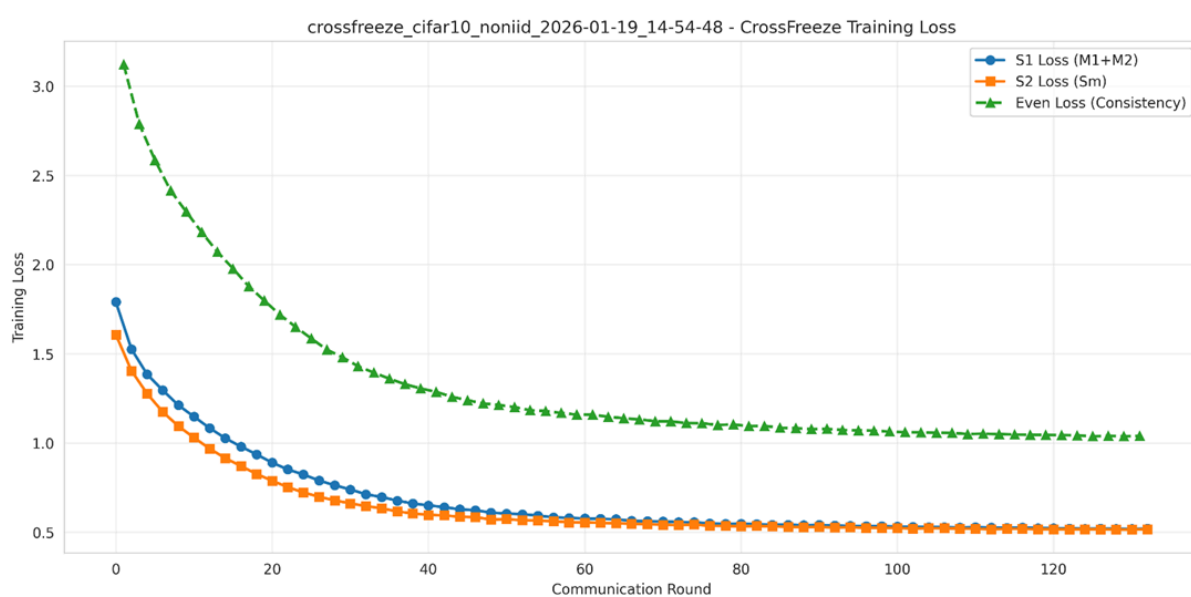
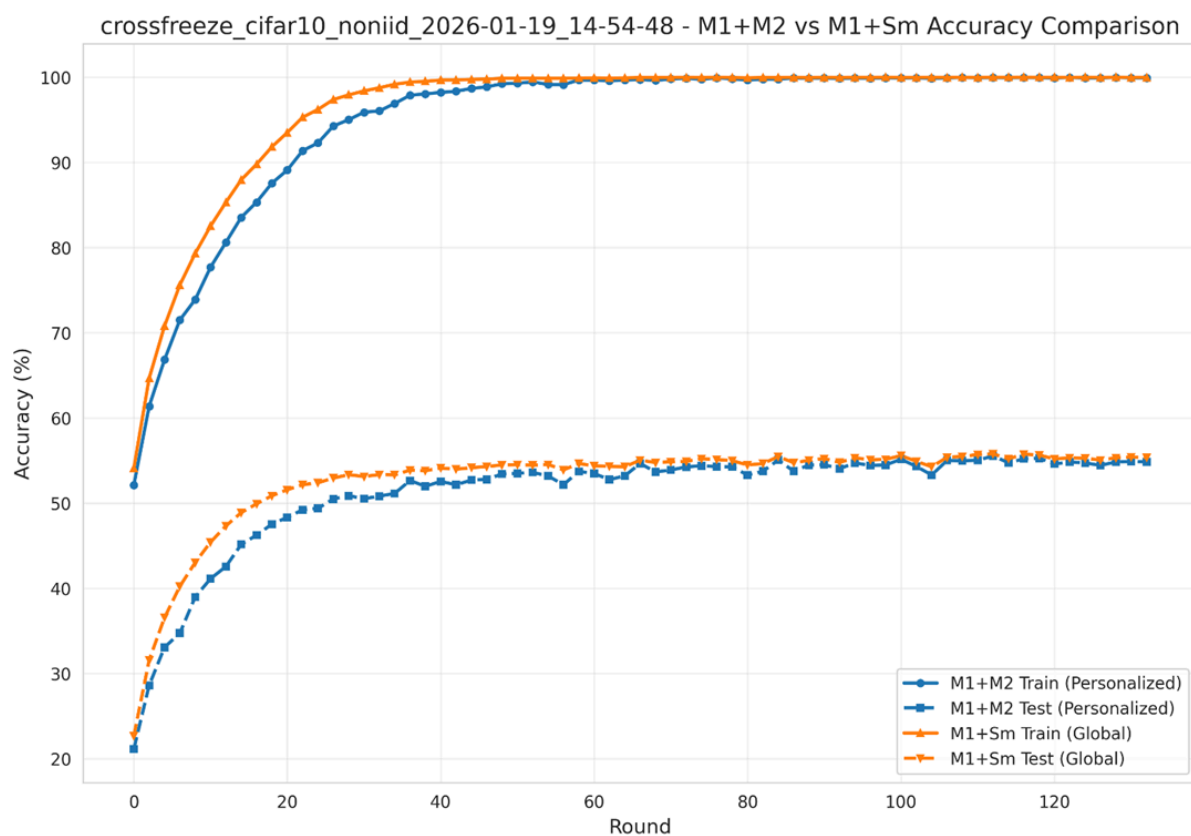
```

● 已停止!
  当前准确率: 55.36%
  最佳准确率: 55.78% (第 57 轮)
  已连续 10 轮无改善 (阈值: 0.01%)
  提前结束训练 (计划 200 轮, 实际 133 轮)
训练进度: 66% | 132/200 [5:24:44<2:47:17, 147.61s/it]

=====
CrossFreeze - 各类别测试准确率摘要
=====
类别      M1+M2(个性化)  M1+Sm(全局)  差值 (M2-Sm)
-----
airplane   54.95%         55.68%        -0.73%
automobile 66.01%         66.37%        -0.35%
bird       52.85%         53.11%        -0.26%
cat        44.74%         45.22%        -0.47%
deer       42.29%         42.29%        -0.00%
dog        45.47%         46.45%        -0.97%
frog       56.77%         57.03%        -0.26%
horse      65.16%         65.89%        -0.73%
ship       59.87%         60.55%        -0.68%
truck      60.35%         61.00%        -0.65%
平均      54.85%         55.36%        -0.51%
=====

指标已保存到: ./results_cifar10_exp_B/crossfreeze_cifar10_noniid_2026-01-19_14-54-48/metrics.json
==== CrossFreeze Summary ====
M1+M2 最终准确率: 54.85%
M1+Sm 最终准确率: 55.36%
M1+M2 最佳准确率: 55.54%
M1+Sm 最佳准确率: 55.78%
生成 CrossFreeze 可视化图表...
CrossFreeze 可视化图表已保存到: ./results_cifar10_exp_B/crossfreeze_cifar10_noniid_2026-01-19_14-54-48
Accuracy curves saved: ./results_cifar10_exp_B/crossfreeze_cifar10_noniid_2026-01-19_14-54-48/accuracy_comparison.png

```



C

- **配置含义:** 在 Exp B 基础上启用 Mixup/CutMix 强增强，验证强增强
- **修改点:** `use_cutmix=1` , `cutmix_alpha=1.0` (恢复强增强), 其他保持 Exp B 配置。

```
python main.py \  
  --experiment crossfreeze \  
  --dataset cifar10 \  
  --gpu 0 \  
  --seed 2024 \  
  --epochs 200 \  
  --num_clients 20 \  
  --frac 1.0 \  
  --local_epochs 5 \  
  --batch_size 64 \  
  --lr 0.02 \  
  --weight_decay 1e-4 \  
  --iid 0 \  
  --alpha 0.5 \  
  --min_lr 0.002 \  
  --gamma_sm 2.0 \  
  --gamma_scheduler static \  
  --consistency_beta 0.0 \  
  --use_cutmix 1 \  
  --mixup_cutmix_ratio 0.5 \  
  --cutmix_alpha 1.0 \  
  --cutmix_prob 0.6 \  
  --save_dir ./results_cifar10_exp_C
```

早停触发!

当前准确率: 57.02%

最佳准确率: 57.37% (第 62 轮)

已连续 10 轮无改善 (阈值: 0.01%)

提前结束训练 (计划 200 轮, 实际 143 轮)

训练进度: 71%

CrossFreeze - 各类别测试准确率摘要

类别	M1+M2(个性化)	M1+Sm(全局)	差值 (M2-Sm)
airplane	54.55%	54.92%	-0.37%
automobile	70.97%	71.97%	-1.00%
bird	51.92%	53.76%	-1.84%
cat	44.32%	45.03%	-0.71%
deer	41.07%	40.26%	0.81%
dog	53.80%	54.19%	-0.38%
frog	59.06%	60.27%	-1.21%
horse	63.85%	66.67%	-2.82%
ship	62.13%	63.03%	-0.90%
truck	59.06%	60.07%	-1.01%
平均	56.07%	57.02%	-0.94%

指标已保存到: ./results_cifar10_exp_C/crossfreeze_cifar10_noniid_2026-01-19_04-59-39/metrics.json

CrossFreeze Summary

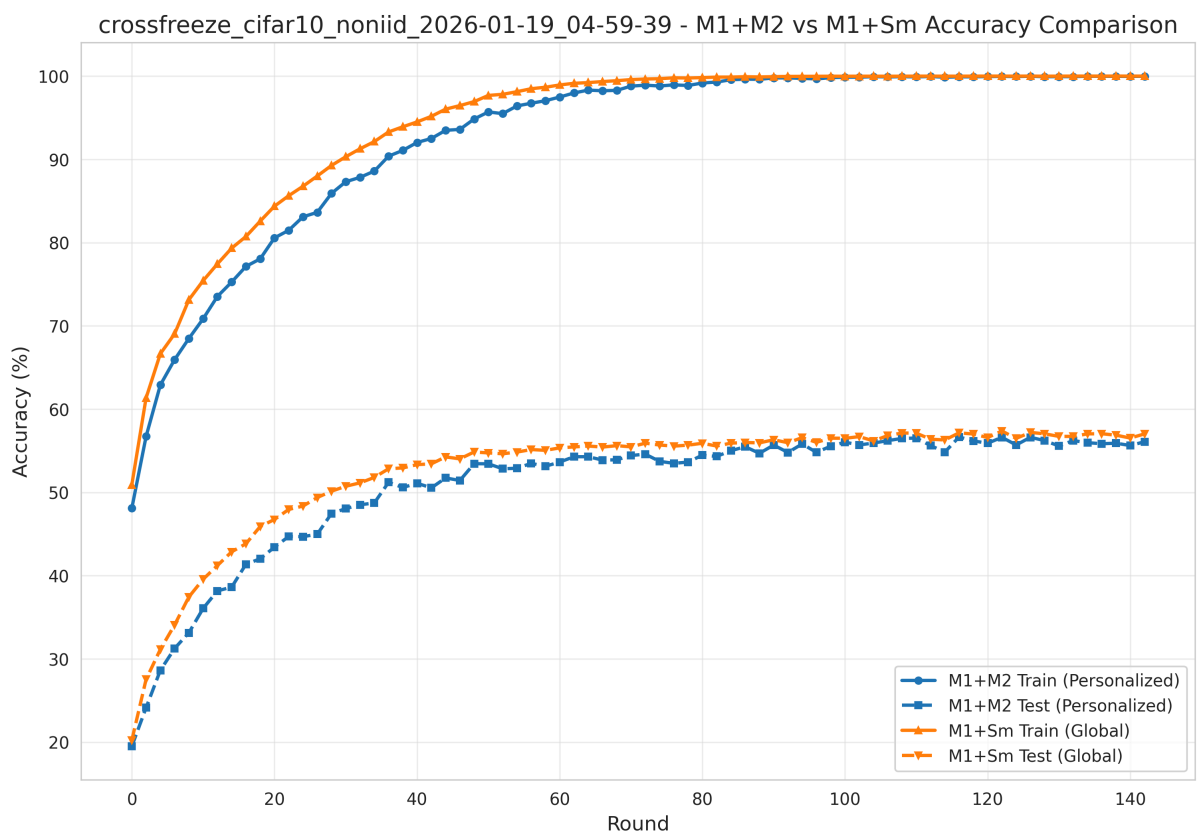
M1+M2 最终准确率: 56.07%

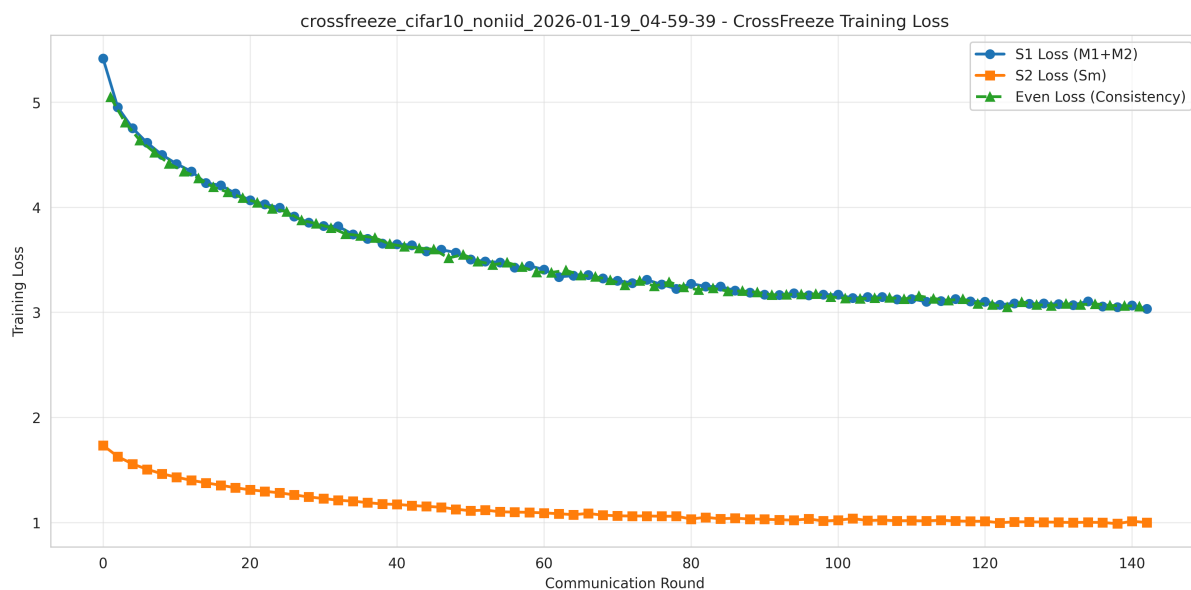
M1+Sm 最终准确率: 57.02%

M1+M2 最佳准确率: 56.65%

M1+Sm 最佳准确率: 57.37%

生成报告: 可视化图表





D

配置含义: 在 Exp C 基础上，强制强弱视图特征一致 (L_{cons})。 **修改点:**

`consistency_beta=1.0` (开启一致性损失), `gamma_scheduler` 仍为 static。

```
python main.py \
  --experiment crossfreeze \
  --dataset cifar10 \
  --gpu 0 \
  --seed 2024 \
  --epochs 200 \
  --num_clients 20 \
  --frac 1.0 \
  --local_epochs 5 \
  --batch_size 64 \
  --lr 0.02 \
  --weight_decay 1e-4 \
  --iid 0 \
  --alpha 0.5 \
  --min_lr 0.002 \
  --gamma_sm 2.0 \
  --gamma_scheduler static \
  --consistency_beta 1.0 \
  --use_cutmix 1 \
  --mixup_cutmix_ratio 0.5 \
```

```
--cutmix_alpha 1.0 \  
--cutmix_prob 0.6 \  
--save_dir ./results_cifar10_exp_D
```

```
早停触发!  
当前准确率: 57.19%  
最佳准确率: 57.50% (第 62 轮)  
已连续 10 轮无改善 (阈值: 0.01%)  
提前结束训练 (计划 200 轮, 实际 143 轮)  
训练进度: 71%|
```

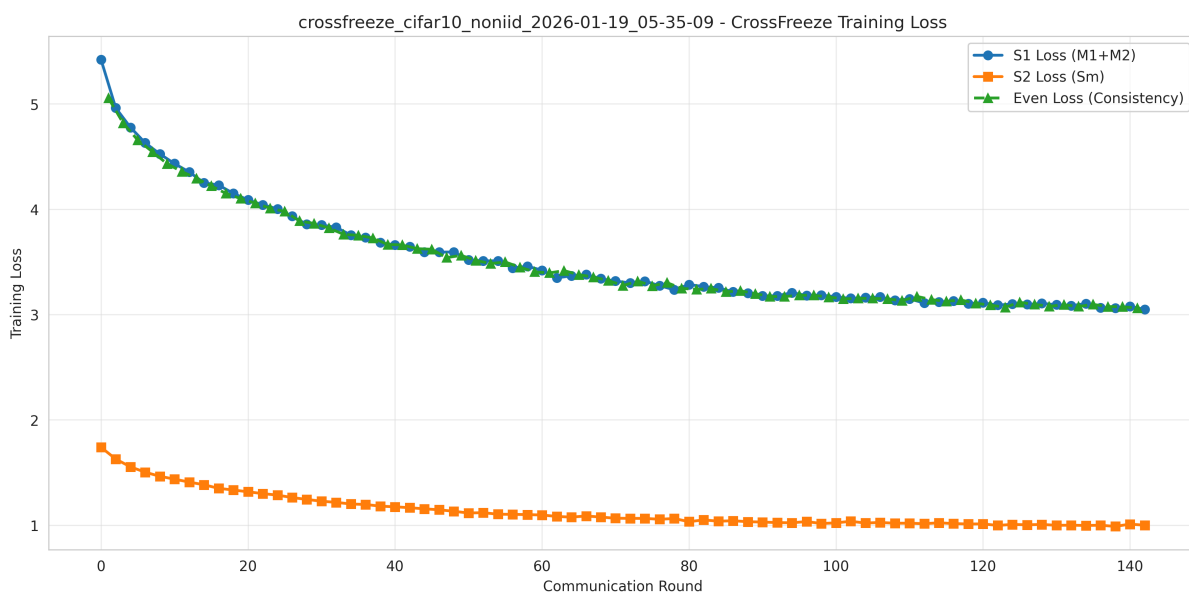
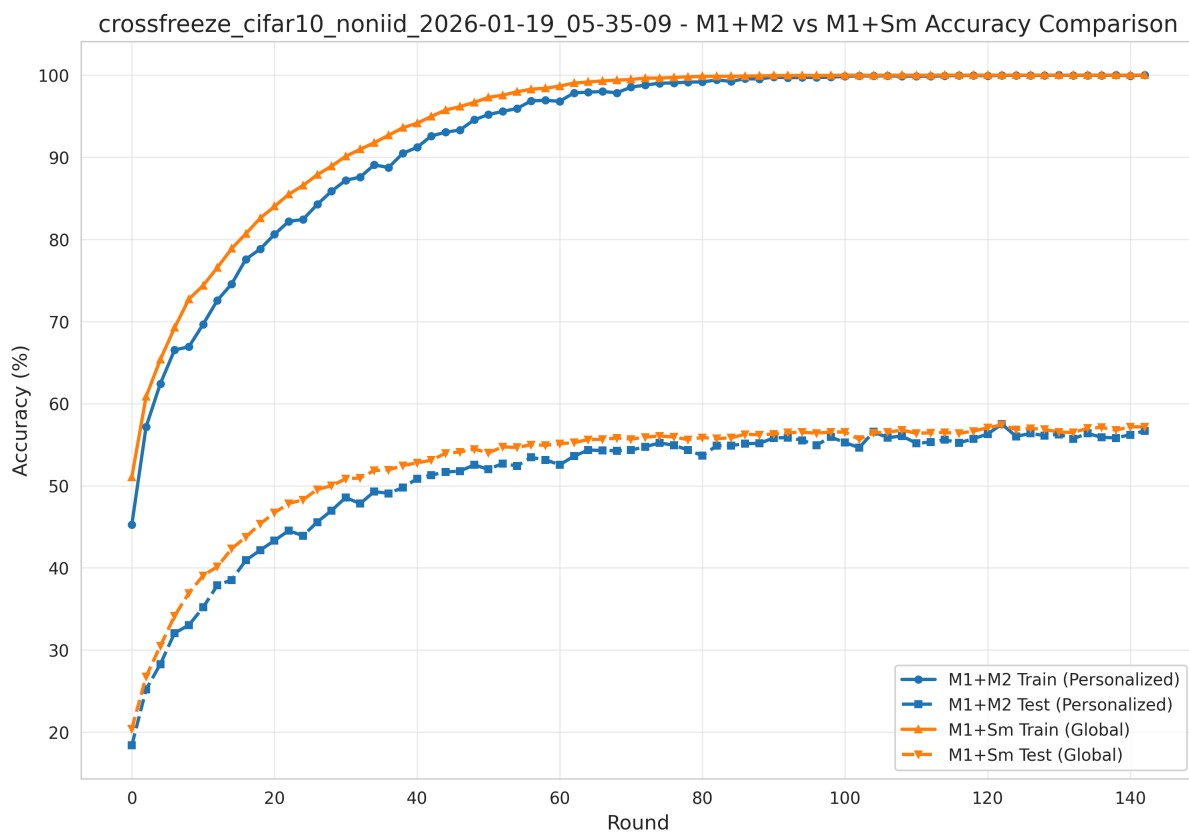
CrossFreeze - 各类别测试准确率摘要

类别	M1+M2 (个性化)	M1+S _m (全局)	差值 (M2-S _m)
airplane	51.64%	53.88%	-2.23%
automobile	70.66%	71.42%	-0.76%
bird	56.98%	55.74%	1.24%
cat	41.74%	43.92%	-2.18%
deer	41.77%	41.29%	0.48%
dog	53.41%	52.85%	0.56%
frog	59.14%	60.61%	-1.48%
horse	67.44%	67.64%	-0.21%
ship	63.37%	63.48%	-0.11%
truck	60.64%	61.02%	-0.39%
平均	56.68%	57.19%	-0.51%

指标已保存到: ./results_cifar10_exp_D/crossfreeze_cifar10_noniid_2026-01-19_05-35-09/metrics.json

== CrossFreeze Summary ==

```
M1+M2 最终准确率: 56.68%  
M1+Sm 最终准确率: 57.19%  
M1+M2 最佳准确率: 57.50%  
M1+Sm 最佳准确率: 57.41%
```



E

- **配置含义:** 完整方案, 使用动态 Soft Warm-up 策略。
- **修改点:** `gamma_scheduler=dynamic` (开启动态权重), 其余参数全开。

```
python main.py \  
  --experiment crossfreeze \  
  --dataset cifar10 \  
  --gpu 0 \  
  --seed 2024 \  
  --epochs 200 \  
  --num_clients 20 \  
  --frac 1.0 \  
  --local_epochs 5 \  
  --batch_size 64 \  
  --lr 0.02 \  
  --weight_decay 1e-4 \  
  --iid 0 \  
  --alpha 0.5 \  
  --min_lr 0.002 \  
  --gamma_sm 2.0 \  
  --gamma_scheduler dynamic \  
  --consistency_beta 1.0 \  
  --use_cutmix 1 \  
  --mixup_cutmix_ratio 0.5 \  
  --cutmix_alpha 1.0 \  
  --cutmix_prob 0.6 \  
  --save_dir ./results_cifar10_exp_E
```

早停触发!
当前准确率: 56.70%
最佳准确率: 56.78% (第 51 轮)
已连续 10 轮无改善 (阈值: 0.01%)
提前结束训练 (计划 200 轮, 实际 121 轮)
训练进度: 60%

CrossFreeze - 各类别测试准确率摘要

类别	M1+M2(个性化)	M1+Sm(全局)	差值(M2-Sm)
airplane	53.52%	54.12%	-0.60%
automobile	69.90%	70.62%	-0.73%
bird	56.52%	55.13%	1.39%
cat	41.72%	44.24%	-2.52%
deer	38.72%	39.20%	-0.49%
dog	51.44%	52.09%	-0.64%
frog	59.36%	60.21%	-0.85%
horse	68.91%	68.19%	0.72%
ship	63.29%	64.00%	-0.71%
truck	58.29%	59.22%	-0.93%
平均	56.17%	56.70%	-0.54%

指标已保存到: ./results_cifar10_exp_E/crossfreeze_cifar10_noniid_2026-01-19_04-47-49/metrics.json

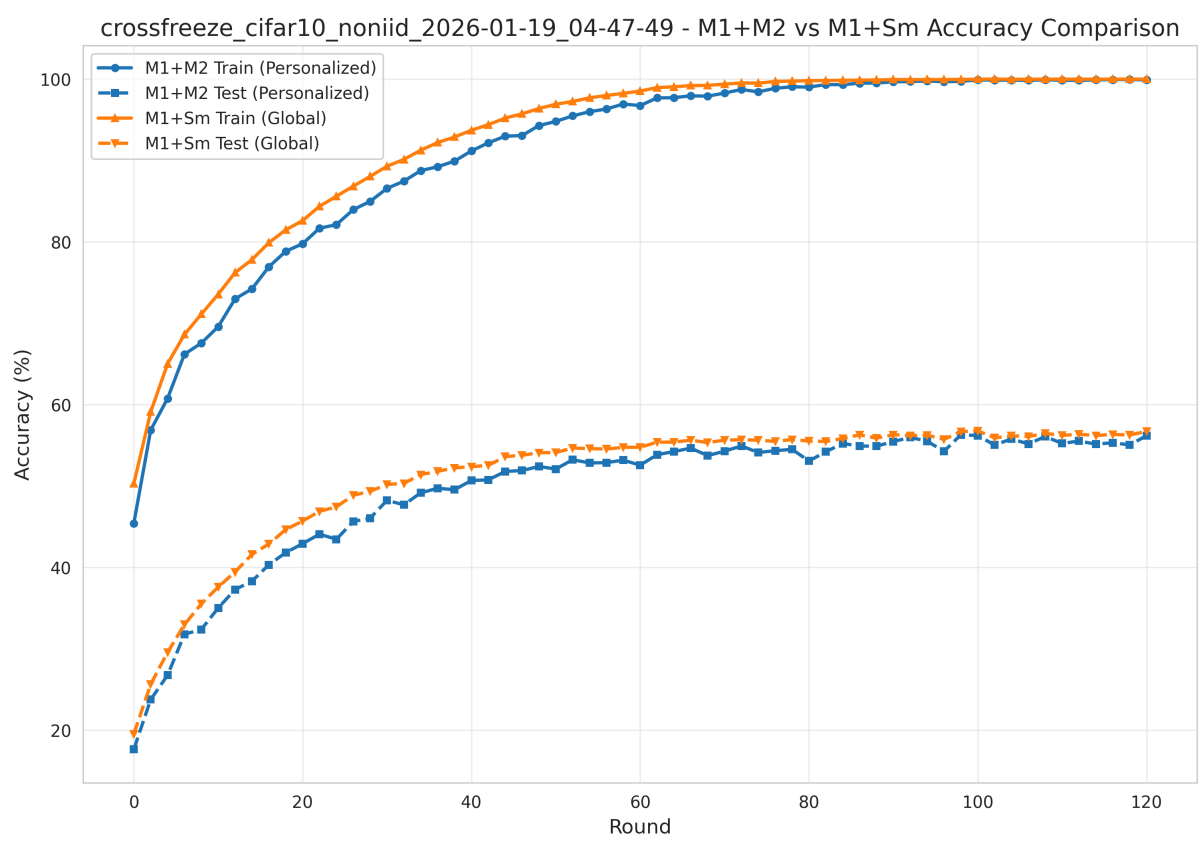
==== CrossFreeze Summary ====

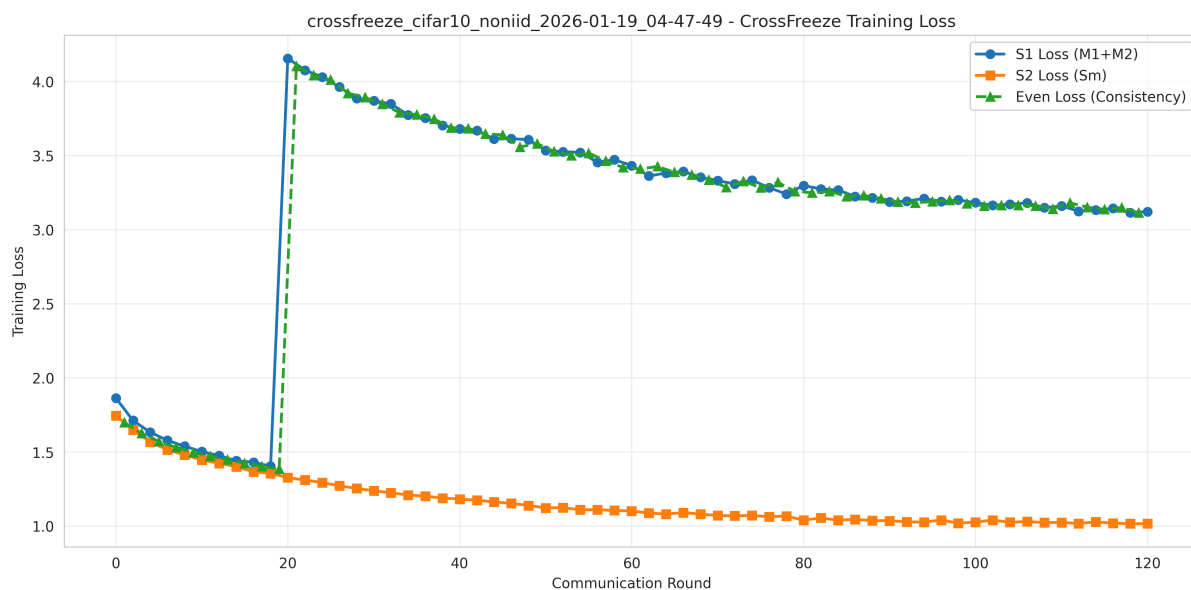
M1+M2 最终准确率: 56.17%

M1+Sm 最终准确率: 56.70%

M1+M2 最佳准确率: 56.28%

M1+Sm 最佳准确率: 56.78%



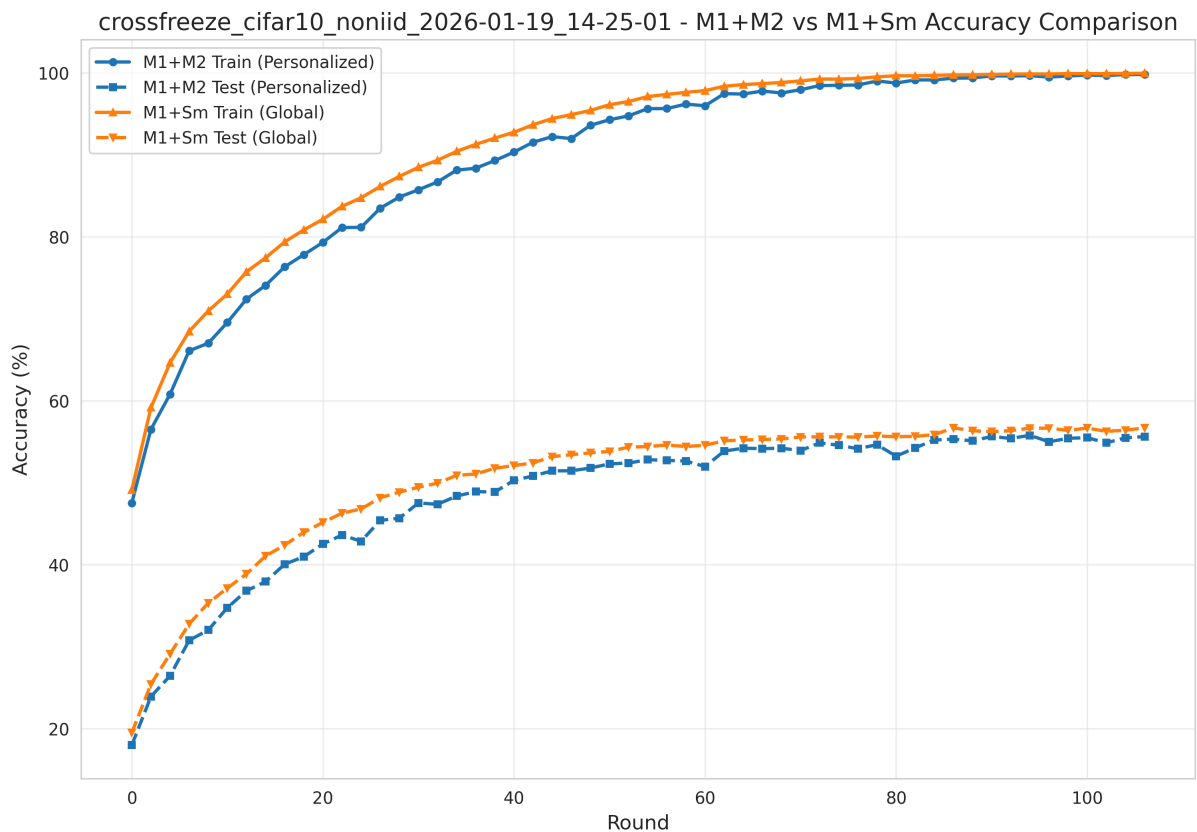
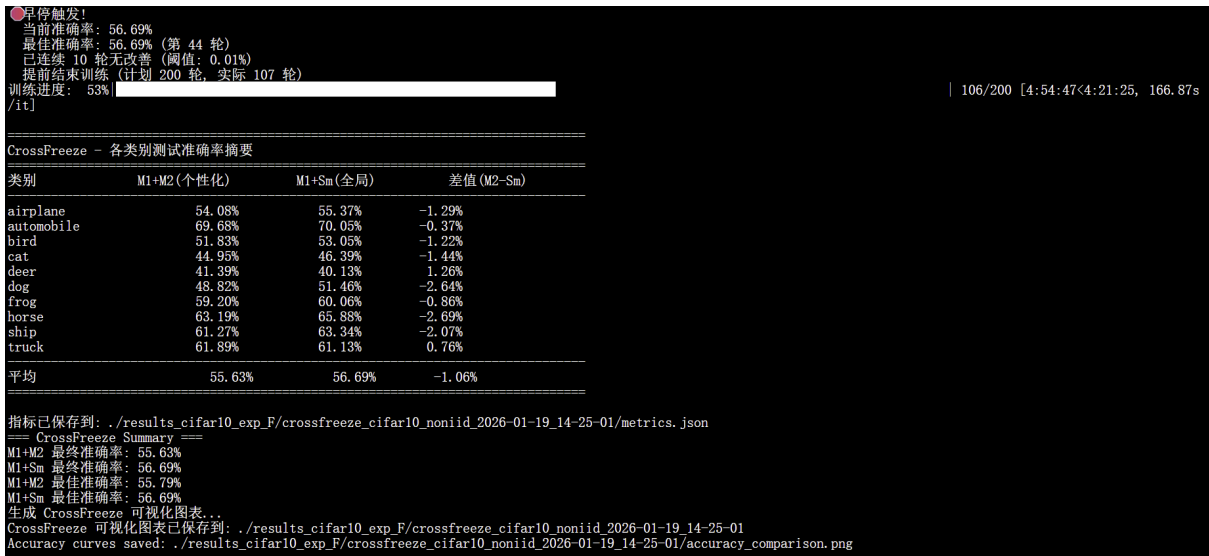


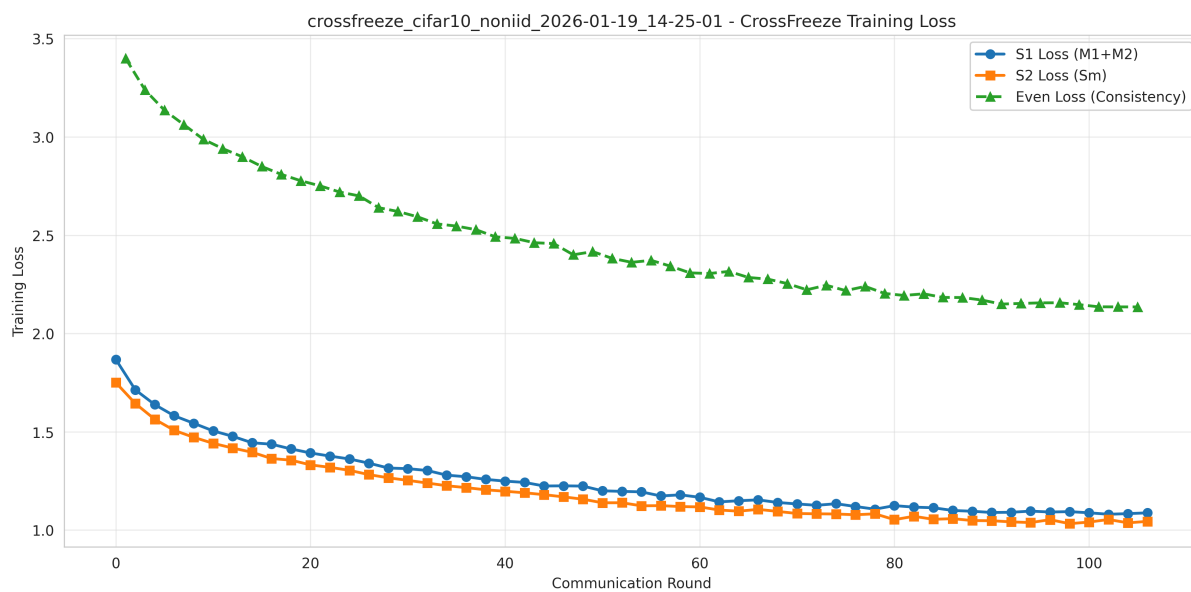
F

验证损失函数范围，在完整的方案基础上损失函数各取所需

```
python main.py \
  --experiment crossfreeze \
  --dataset cifar10 \
  --gpu 0 \
  --seed 2024 \
  --epochs 200 \
  --num_clients 20 \
  --frac 1.0 \
  --local_epochs 5 \
  --batch_size 64 \
  --lr 0.02 \
  --weight_decay 1e-4 \
  --iid 0 \
  --alpha 0.5 \
  --min_lr 0.002 \
  --gamma_sm 2.0 \
  --gamma_scheduler static \
  --consistency_beta 1.0 \
  --use_cutmix 1 \
  --mixup_cutmix_ratio 0.5 \
  --cutmix_alpha 1.0 \
```

```
--cutmix_prob 0.6 \
--separate_loss 1 \
--save_dir ./results_cifar10_exp_F
```

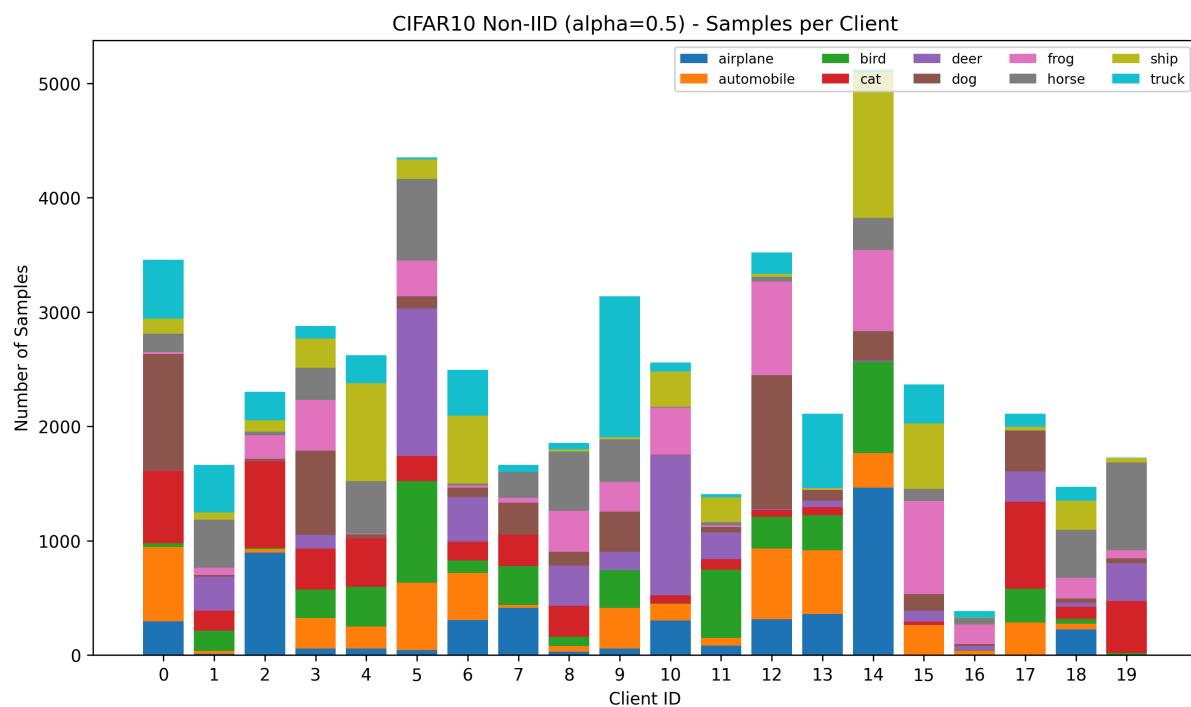


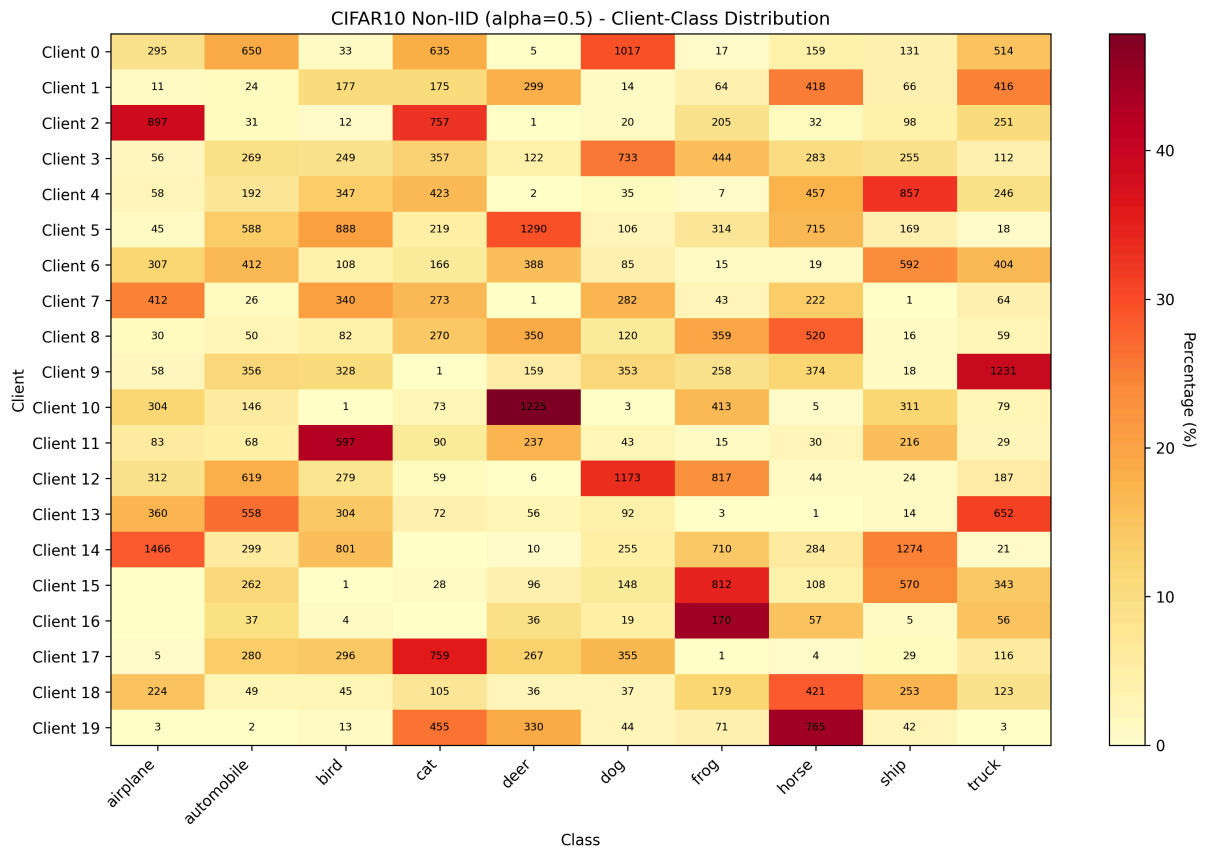


ood可视化

输入

```
python visualize_data_distribution.py --dataset cifar10 --num_clients 20 --alpha 0.5
```







实验结论

我感觉各个模块的增益被抵消了。

目前的数据显示**强数据增强 (Strong Aug)** 带来了显著提升 (+1.29%)，但 **一致性正则 (Consistency)** 几乎无效 (+0.04%)，而 **动态策略 (Dynamic Scheduler)** 甚至带来了负优化 (-0.63%)。